電気回路Ⅲ　配布資料３

・アドミタンスの定義

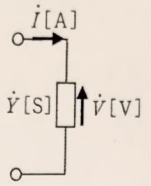


図 5-43

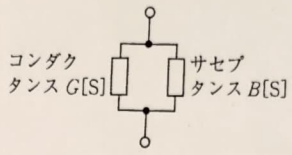
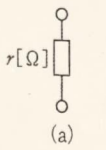


図 5-44　$\dot{Y}=G+jB$[S]が表す回路

・回路要素のアドミタンス



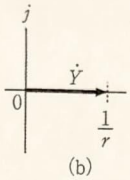
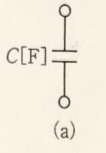


図 5-46



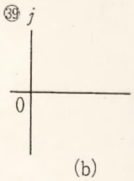
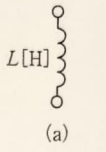


図 5-48



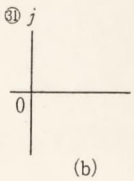


図 5-47

回路要素	アドミタンス[S]	コンダクタンス[S]	サセプタンス[S]
抵抗r[Ω]			
インダクタンスL[H]			
キャパシタンスC[F]			

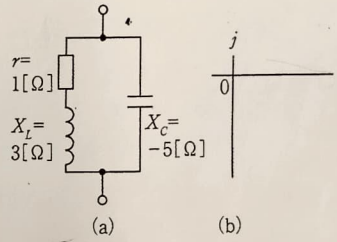
問題１　次の回路のアドミタンス$\dot{Y}$ [S] を求めよ，そのG [S]，B [S]，アドミタンスの絶対値$|\dot{Y}|$ [S]，θ_Y [rad] を求めよ．

$\dot{Y} = (1+j3)^{-1} + \left(-\frac{1}{5}j\right)$

$= \frac{1}{1+j3} + \frac{j}{5}$ [S]

$= \frac{1-j3}{1+9} + \frac{j}{5} = \frac{1-j}{10}$ [S]

$G = \frac{1}{10}$ [S]，$B = \frac{-1}{10}$ [S]



$|\dot{Y}| = \frac{\sqrt{2}}{10}$ [S]

$\theta_Y = -45°$

問図 5-31

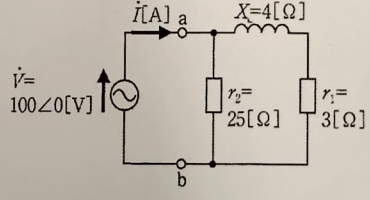
問題２　図の回路に流れる電流i [A] の極座標表示を求めよ．

$\dot{X} = \left(25^{-1} + (3+j4)^{-1}\right)^{-1} = \left(\frac{3-j4}{25} + \frac{1}{25}\right)^{-1}$

$= \frac{25}{4-j4}$

$= \frac{25(4+j4)}{32}$

$100 \times \frac{4-j4}{25} = 16-j16 = 16\sqrt{2}\exp\left(-j\frac{\pi}{4}\right)$



問図 5-39